PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

JACKFRUIT – MINIPROJECT

SOFTWARE ENGINEERING – PROJECT

SOFTWARE REQUIREMENTS SPECIFICATION

Priority-Based Medicine Booking and Allocation Module for Pharma Apps

Version 1.0
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Project ID: 47

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Revision History

Name	Date	Reason For Changes	Version
Project Team	05-09-2026	Initial SRS preparation	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the Priority-Based Medicine Booking and Allocation Module for Pharma Apps. The module is intended to support medicine booking by considering medicine availability, booking urgency, and relevant patient history. It provides controlled allocation of available stock, manages waitlists when stock is unavailable, sends status notifications, and supports integration with electronic prescriptions (e-prescriptions).

This document specifies the functional and nonfunctional requirements that will guide analysis, architecture, design, implementation, testing, validation, and maintenance.

1.2 Intended Audience and Reading Suggestions

This SRS is intended for:

- Project team members responsible for development and implementation.
- Faculty evaluators and project reviewers.
- Software architects and designers preparing UML, database, and interface designs.
- Test engineers preparing unit, integration, and system test cases.
- Pharmacy staff or administrators who will manage inventory and fulfil bookings.
- Users/customers who book medicines through the pharmacy application.

Readers should first review Sections 1 and 2 for scope and context, Section 3 for interfaces, Section 4 for analysis models, Section 5 for detailed functional requirements, Section 6 for quality and security requirements, and the appendices for field layouts and requirement traceability.

1.3 Product Scope

The proposed system is a pharmacy-app module that manages medicine booking and allocation using a priority score. A booking can be prioritised using configurable factors such as urgency, medicine availability, and permitted patient-history information. When sufficient stock exists, the system allocates stock according to the priority order and booking rules. When stock is unavailable, eligible bookings are placed on a waitlist. The system automatically updates users when stock becomes available or when booking status changes.

- User registration, login, and role-based access.
- Medicine search and current availability display.
- E-prescription upload/verification and prescription-linked booking.
- Priority calculation and priority-ordered booking allocation.
- Waitlist creation, ordering, promotion, and cancellation.
- Notifications for booking, allocation, waitlist, prescription, and fulfilment events.
- Pharmacy inventory updates and order fulfilment status tracking.
- Administrative dashboards, audit logs, and reporting.

The system is a booking and allocation aid; it does not diagnose diseases, prescribe medicines, or independently change a doctor's prescription.

1.4 References

- PES University, Department of Computer Science and Engineering, JACKFRUIT – Software Engineering Project Guidelines, UE24CS341A, 2026.
- Project topic specification: Priority-Based Medicine Booking and Allocation Module for Pharma Apps, Project ID 47.
- Relevant e-prescription/API documentation selected during implementation.
- Applicable organizational pharmacy policies and data-protection requirements identified during system design.

2. Overall Description

2.1 Product Perspective

The module is designed as a subsystem of a web or mobile pharmacy application. It interacts with customers, pharmacy staff, administrators, an application database, inventory services, notification services, and an e-prescription component.

Component	Purpose
Customer Application	Search medicines, upload/select prescription, request booking, view status and waitlist position.
Booking & Priority Engine	Validates bookings, calculates configurable priority, orders requests, and allocates available stock.
Inventory Service	Maintains medicine stock and reserves/releases quantities.
Prescription Service	Stores and validates prescription-linked information required for eligible medicines.
Notification Service	Sends booking, allocation, waitlist, prescription and fulfilment notifications.
Pharmacy Staff Interface	Reviews bookings, manages inventory, validates fulfilment and handles exceptions.
Admin Module	Manages users, medicine master data, rules, audit records and reports.
Database	Stores users, medicines, prescriptions, bookings, priority factors, waitlists, notifications and audit data.

2.2 Product Functions

- Authenticate users and enforce role-based access.
- Search medicines by name/category and view availability.
- Accept booking requests with required quantity and relevant prescription details.
- Calculate and assign a booking priority using defined rules.
- Allocate stock to the highest eligible priority requests when stock becomes available.
- Place unfulfilled requests on a waitlist and maintain their order.
- Automatically promote eligible waitlisted requests when inventory is replenished.
- Send notifications when booking, prescription, allocation, waitlist, or fulfilment status changes.

- Allow pharmacy staff to update inventory and process bookings.
- Maintain audit trails for important actions and status changes.
- Provide administrative reports on bookings, allocation, waitlists, inventory, prescription status, notifications and audit records.

2.3 User Classes and Characteristics

User Class	Characteristics	Main Permissions
Customer/Patient	Basic smartphone/web skills; interacts with booking and notification features.	Search, request booking, upload/view prescription, view status, cancel eligible booking.
Pharmacy Staff	Trained pharmacy/application operator.	View/validate bookings, manage inventory, allocate/fulfil orders, manage waitlist exceptions.
Administrator	Technical/business administrator.	Manage users, roles, medicine records, priority rules, reports and audit logs.
System/Integration Service	Machine-to-machine component.	Exchange inventory, prescription and notification data through authenticated interfaces.

2.4 Operating Environment

- Client: modern web browser and/or Android mobile application.
- Server: application server capable of running the selected backend framework.
- Database: relational database such as MySQL/PostgreSQL.
- Network: HTTPS-enabled network connectivity.
- Development: Git/GitHub, CI/CD pipeline, Docker, automated testing and static analysis tools as required by the course guidelines.
- Notification infrastructure: email/SMS/push notification service selected for implementation.

2.5 Design and Implementation Constraints

- The system shall follow Agile development practices as required by the course guidelines.
- GitHub shall be used for version control and team contribution tracking.
- Role-based access control is required because the system handles user and prescription-related information.
- Priority rules must be deterministic, documented, testable and configurable by authorized administrators.
- Prescription information shall not be modified by the booking module.
- Only authorized pharmacy personnel shall perform inventory and fulfilment operations.
- The implementation must support unit, integration and system testing.
- External services may impose API availability, authentication, rate and payload constraints.

2.6 Assumptions and Dependencies

- Users provide accurate booking and prescription information.
- Medicine inventory quantities are updated by authorized pharmacy personnel or an integrated inventory service.
- Prescription verification, where required, is performed by an authorized workflow.
- Notification delivery depends on the availability of the selected notification provider.

- External e-prescription systems expose an interface or data format that can be integrated with the module.
- The priority policy is supplied by the project/customer and can be changed only by authorized personnel.
- The prototype may use simulated prescription and inventory services if real external APIs are not available.

3. External Interface Requirements

3.1 User Interfaces

- Login/registration screen with role-appropriate navigation.
- Medicine search screen showing medicine name, strength/form where applicable, price if implemented, and available quantity/status.
- Booking screen showing requested quantity, prescription requirement/status, urgency selection, and booking confirmation.
- Booking status screen showing priority/status, allocation result, waitlist position if applicable, and relevant timestamps.
- Pharmacy dashboard showing pending bookings, stock alerts, waitlist and fulfilment actions.
- Administration dashboard showing medicine records, priority-rule configuration, user/role controls, reports and audit information.
- Notification center showing unread and historical notifications.
- Validation and error messages shall clearly identify invalid inputs and required corrective actions.

3.2 Software Interfaces

Interface	Input	Output/Purpose
Application Database	User, medicine, inventory, booking, prescription, priority-rule, waitlist, notification and audit data	Persistent storage and retrieval.
E-Prescription Interface	Prescription identifier/document/structured prescription data	Prescription status and required medicine information.
Inventory Interface	Medicine ID and quantity adjustments	Current stock, reservation and release status.
Notification Interface	Recipient, message, event type	Delivery status for email/SMS/push notification.
Authentication Component	Credentials/session/token	Authenticated user identity and role.

3.3 Communications Interfaces

- Client-server communication shall use HTTPS.
- REST/JSON may be used for application APIs unless another approved interface is selected.
- Authenticated API requests shall use secure session/token mechanisms.
- External notification and prescription services shall use their documented authenticated protocols.
- Communication failures shall be logged and shall not silently change booking or inventory state.

4. Analysis Models

The following analysis models define the functional interactions, workflow, system structure, data relationships, and major data flows of the proposed system.Use Case Diagram – Customer, Pharmacy Staff and Administrator interactions.

- Activity Diagram – End-to-end priority booking and allocation workflow.
- Sequence Diagram – Booking request through priority calculation, inventory check, allocation/waitlist and notification.
- Class Diagram – User, Medicine, Prescription, Booking, PriorityRule, WaitlistEntry, Inventory, Notification and AuditLog.
- Entity-Relationship Diagram – Database entities and relationships.
- Data Flow Diagram – Customer application, booking engine, inventory, prescription and notification components.

4.1 Use Case Diagram

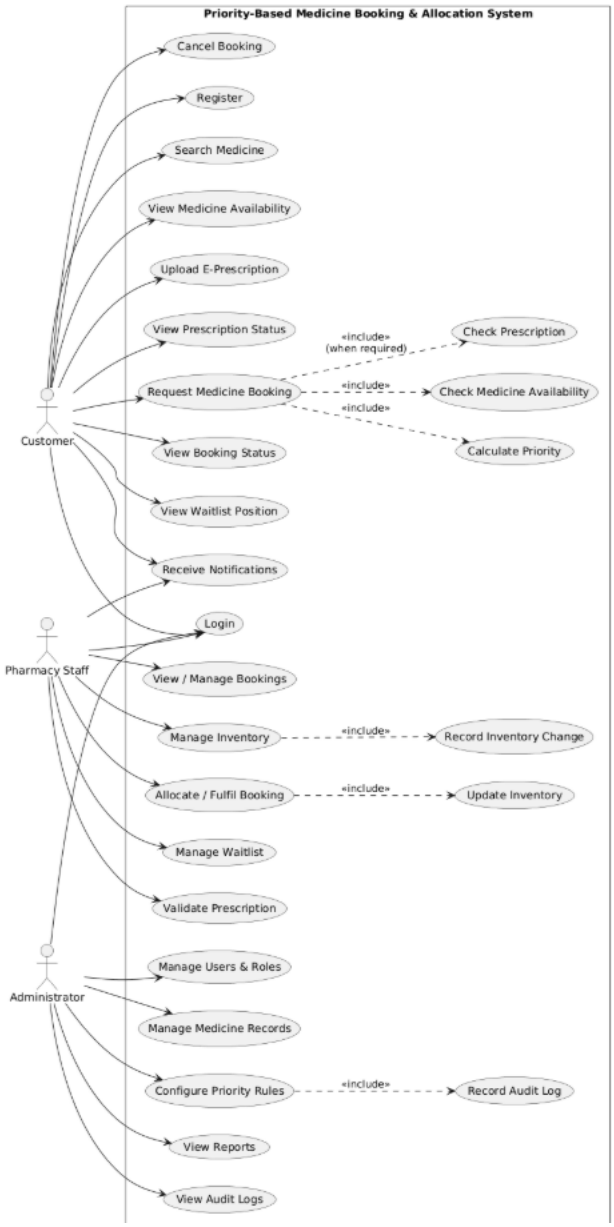


Figure 4.1. Use Case Diagram

4.2 Activity Diagram

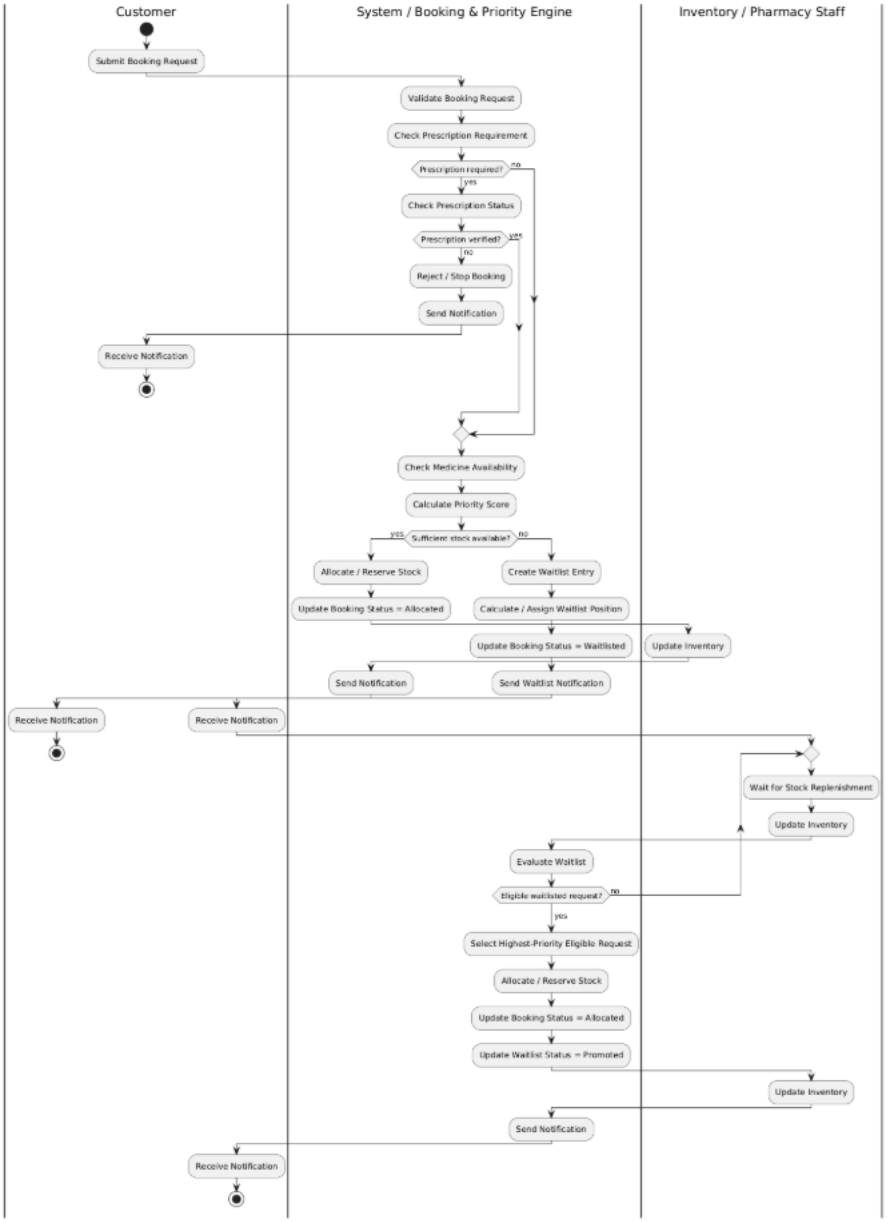


Figure 4.2. Activity Diagram – Priority-Based Booking and Allocation

4.3 Sequence Diagram

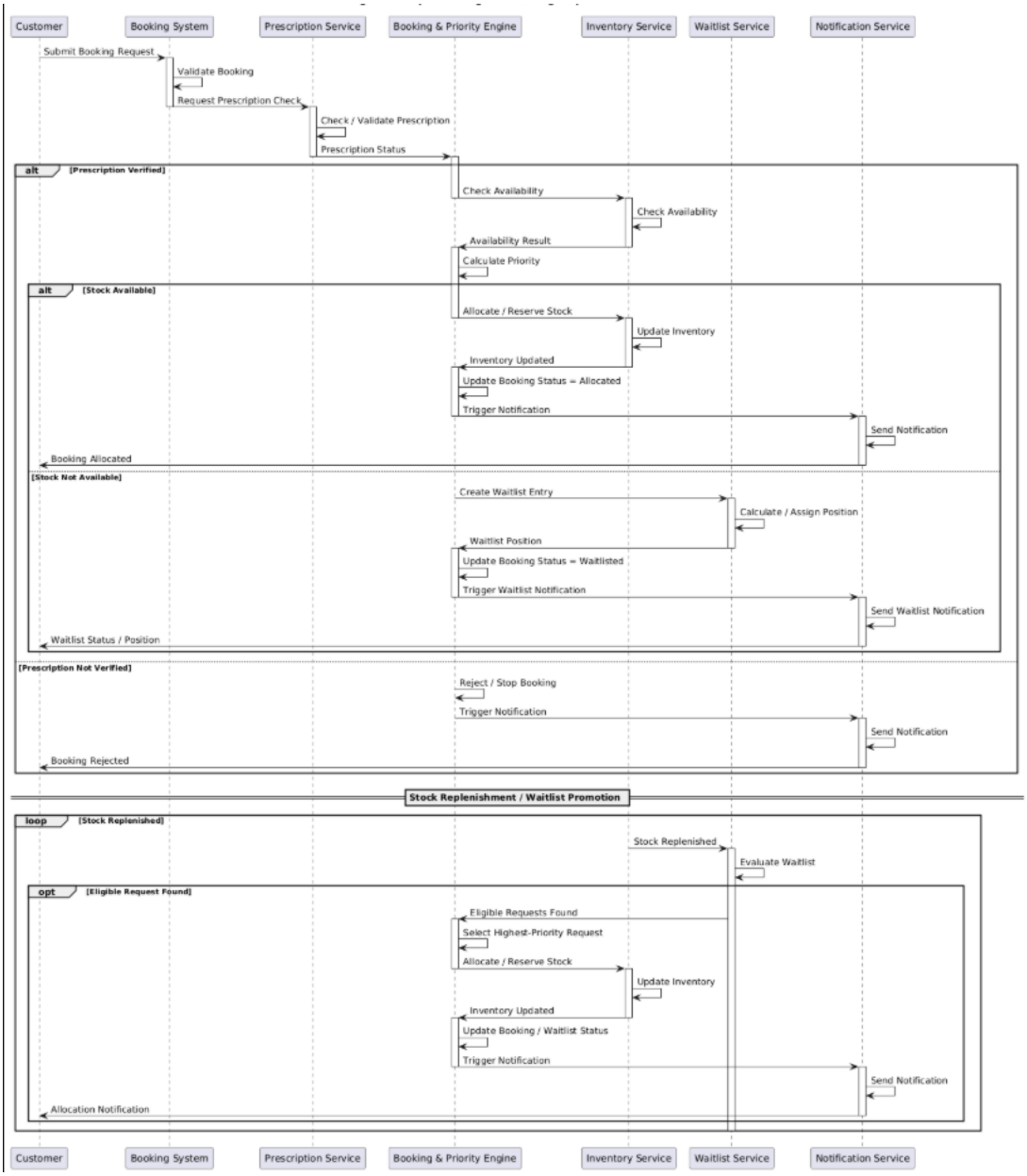


Figure 4.3. Sequence Diagram – Booking Request and Allocation

4.4 Class Diagram

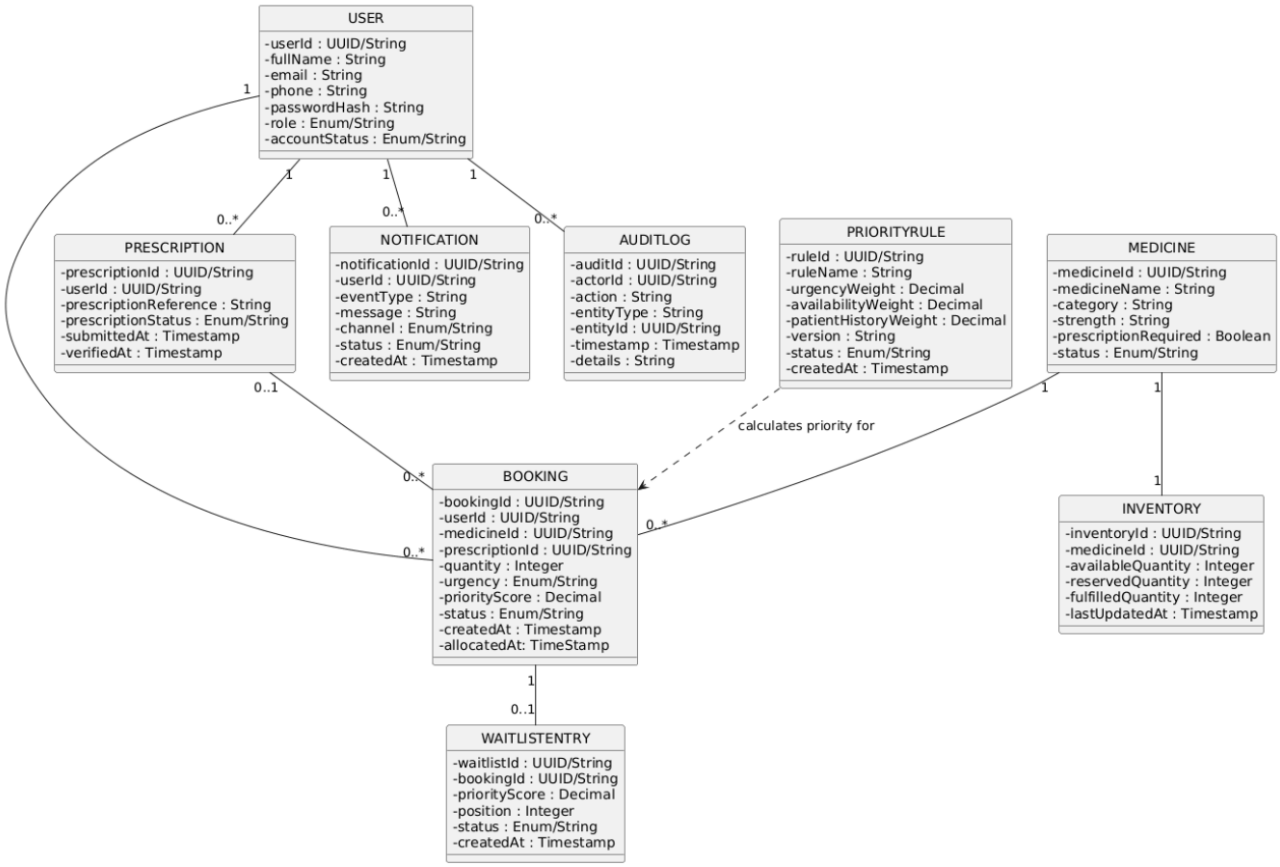


Figure 4.4. Class Diagram

4.5 Entity-Relationship Diagram

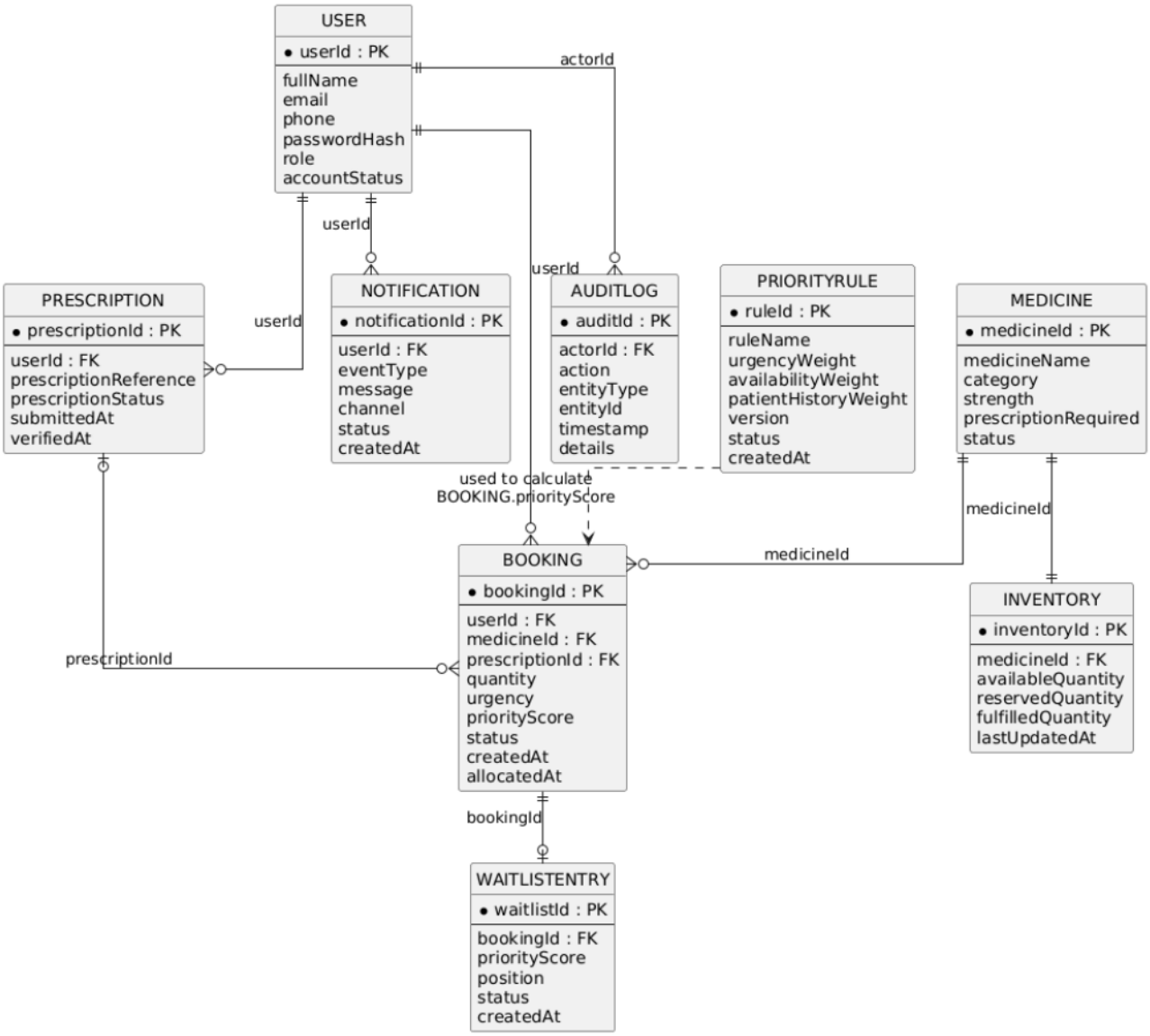


Figure 4.5. Entity-Relationship Diagram

4.6 Data Flow Diagram

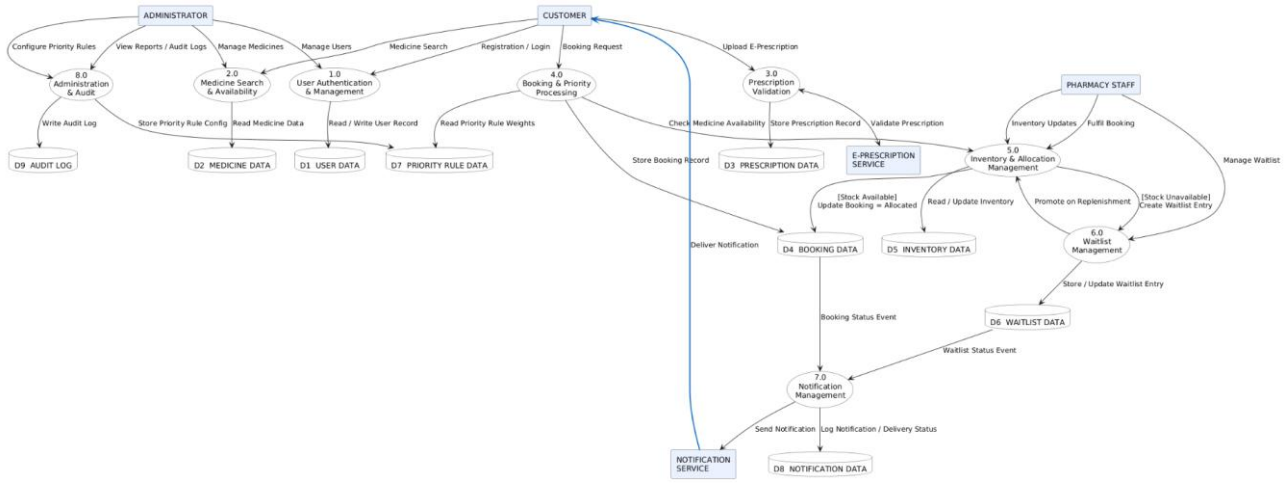


Figure 4.6. Data Flow Diagram

5. System Features

5.1 User Registration and Authentication

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

1. User enters credentials → system validates credentials → system creates authenticated session or rejects the request.

Functional Requirements

Requirement ID	Requirement
FR-AUTH-01	The system shall allow a customer to create an account using required registration fields.
FR-AUTH-02	The system shall authenticate users before allowing access to protected booking functions.
FR-AUTH-03	The system shall assign and enforce user roles such as Customer, Pharmacy Staff and Administrator.
FR-AUTH-04	The system shall deny unauthorized access to role-restricted operations.
FR-AUTH-05	The system shall provide appropriate error messages for invalid login credentials.

5.2 Medicine Search and Availability

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

2. User searches medicine → system retrieves matching medicines and availability → system displays results.

Functional Requirements

Requirement ID	Requirement
FR-MED-01	The system shall allow customers to search medicines using supported search criteria.
FR-MED-02	The system shall display current availability status for a selected medicine.
FR-MED-03	The system shall prevent a booking quantity from exceeding allocatable stock unless the request is eligible for waitlisting.
FR-MED-04	The system shall allow authorized staff to add, update and deactivate medicine records.
FR-MED-05	The system shall record inventory changes with timestamp and responsible user/service.

5.3 E-Prescription Integration

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

3. User submits prescription → system validates required information → system records prescription status → booking proceeds only when applicable conditions are satisfied.

Functional Requirements

Requirement ID	Requirement
FR-PRES-01	The system shall allow a customer to attach an e-prescription to a booking when required.
FR-PRES-02	The system shall validate that required prescription information is present before submitting a prescription-dependent booking.
FR-PRES-03	The system shall preserve prescription information without modifying the original prescribed medicine or dosage.
FR-PRES-04	The system shall show prescription status such as Submitted, Under Review, Verified or Rejected.
FR-PRES-05	The system shall prevent fulfilment of a prescription-dependent booking when required verification has not been completed.

5.4 Priority-Based Booking and Allocation

Priority Calculation

The system shall calculate a normalized priority score using configurable weights for urgency, medicine availability, and permitted patient-history factors. The exact weights and factor ranges shall be defined and approved during the design phase. Eligible booking requests shall be ordered in descending priority score. If two or more requests have the same priority score, the earlier booking timestamp shall be used

as the tie-breaker. The priority calculation shall not be used for diagnosis, treatment recommendation, or modification of a prescription.

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

Booking is submitted → system validates request → system checks stock→ system calculates priority → system allocates stock or triggers waitlist→ system records status and notifies affected users.

Functional Requirements

Requirement ID	Requirement
FR-PRI-01	The system shall create a booking request containing medicine, quantity, user, timestamp and applicable urgency information.
FR-PRI-02	The system shall calculate a priority value using configured factors including urgency, medicine availability and permitted patient-history information.
FR-PRI-03	The priority calculation shall use a documented and deterministic rule set.
FR-PRI-04	When multiple eligible requests compete for limited stock, the system shall process them in descending priority order.
FR-PRI-05	The system shall use booking timestamp as a tie-breaker when two eligible requests have equal priority, unless an approved alternative rule is configured.
FR-PRI-06	The system shall reserve allocated quantity so that the same stock is not allocated to multiple bookings.
FR-PRI-07	The system shall update booking status after successful allocation.
FR-PRI-08	The system shall record the factors/rule version used to calculate the priority for auditability.

5.5 Waitlist Management

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

4. No sufficient stock → system creates waitlist entry → stock changes → system evaluates waitlist → eligible request is promoted/allocated → notification is generated.

Functional Requirements

Requirement ID	Requirement
FR-WAIT-01	If a valid booking cannot be allocated because sufficient stock is unavailable, the system shall offer/perform waitlist placement according to configured rules.

FR-WAIT-02	The system shall maintain a waitlist position for each eligible request.
FR-WAIT-03	The system shall reorder/promote waitlist entries according to the configured priority policy.
FR-WAIT-04	When stock is replenished, the system shall identify eligible waitlisted requests for allocation.
FR-WAIT-05	The system shall notify an affected customer when their waitlist status changes.
FR-WAIT-06	The system shall allow cancellation of an eligible waitlist request and record the cancellation.

5.6 Notifications

Description and Priority

Priority: Medium. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

5. Booking/status event occurs → system creates notification → notification service attempts delivery → delivery status is stored.

Functional Requirements

Requirement ID	Requirement
FR-NOT-01	The system shall generate a notification when a booking is created.
FR-NOT-02	The system shall notify the customer when a booking is allocated, waitlisted, rejected or cancelled.
FR-NOT-03	The system shall notify relevant staff when action is required for a booking or prescription.
FR-NOT-04	The system shall record notification status such as Pending, Sent, Delivered where supported, or Failed.
FR-NOT-05	The system shall retry failed notifications according to a configurable retry policy.

5.7 Pharmacy Inventory and Order Fulfilment

Description and Priority

Priority: High. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

6. Staff updates stock or booking is allocated → system validates quantity → inventory is updated/reserved/released → audit event is recorded.

Functional Requirements

Requirement ID	Requirement
FR-INV-01	Authorized pharmacy staff shall be able to record stock additions and deductions.
FR-INV-02	The system shall maintain available, reserved and fulfilled quantities.

FR-INV-03	The system shall prevent inventory quantities from becoming negative through normal booking operations.
FR-INV-04	Staff shall be able to move an allocated booking through fulfilment statuses.
FR-INV-05	The system shall release reserved quantity when an eligible booking is cancelled before fulfilment.

5.8 Admin and Audit Management

Description and Priority

Priority: Medium. This feature is necessary to support the core pharmacy booking and allocation workflow.

Stimulus/Response Sequences

7. Administrator requests an administrative operation → system checks role → operation is performed → audit entry is created.

Functional Requirements

Requirement ID	Requirement
FR-ADM-01	Administrators shall be able to manage supported user roles.
FR-ADM-02	Administrators shall be able to configure supported priority rules.
FR-ADM-03	The system shall retain an audit record for priority-rule changes, inventory adjustments and important booking status changes.
FR-ADM-04	Administrators shall be able to view summary reports for bookings, allocation, waitlists and inventory.
FR-ADM-05	Audit records shall identify the action, actor/service, timestamp and affected record where applicable.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- NFR-PERF-01: Normal medicine search requests shall return results within 2 seconds under the agreed prototype test load.
- NFR-PERF-02: A valid booking request shall receive an acknowledgement within 3 seconds under normal load, excluding delays caused by external services.
- NFR-PERF-03: Allocation processing shall prevent duplicate allocation of the same stock quantity under concurrent booking requests.
- NFR-PERF-04: The system shall support the agreed project test workload without unacceptable degradation.

6.2 Safety Requirements

- NFR-SAFE-01: The system shall not independently diagnose a patient or create/modify a medical prescription.
- NFR-SAFE-02: Prescription-dependent bookings shall not be fulfilled when mandatory verification is incomplete.
- NFR-SAFE-03: Inventory allocation shall use atomic/transaction-safe operations to avoid allocating unavailable stock.
- NFR-SAFE-04: Critical booking and inventory failures shall produce an explicit error state rather than silently confirming the transaction.

6.3 Security Requirements

- NFR-SEC-01: Protected functions shall require authentication.
- NFR-SEC-02: Role-based authorization shall restrict staff and administrator functions.
- NFR-SEC-03: Passwords shall never be stored in plaintext; secure password hashing shall be used if local authentication is implemented.
- NFR-SEC-04: Sensitive data shall be transmitted using HTTPS/TLS.
- NFR-SEC-05: Prescription-related information shall be accessible only to authorized users/services.
- NFR-SEC-06: The system shall validate and sanitize user inputs to reduce injection and malformed-input risks.
- NFR-SEC-07: Important security and business events shall be logged without unnecessarily exposing sensitive data.
- NFR-SEC-08: Sessions/tokens shall expire or be invalidated according to the selected authentication design.

6.4 Software Quality Attributes

Attribute	Requirement
Usability	Common booking operations shall be understandable without technical training and shall provide clear status messages.
Reliability	The system shall preserve booking and inventory consistency during expected failures.
Maintainability	Modules shall have clear interfaces and separation of concerns so that priority rules, notifications and inventory services can be changed independently.
Testability	Functional requirements shall be independently testable and mapped to test cases.
Scalability	The design shall allow additional pharmacies, medicines and users without redesigning the core priority workflow.
Interoperability	The system shall use documented APIs/data formats for e-prescription, inventory and notification integrations.
Auditability	Priority decisions and important booking/inventory changes shall be traceable.
Portability	The application shall be deployable using containerization where practical.

6.5 Business Rules

- BR-01: Only authenticated users may create or view their own bookings.

- BR-02: Only authorized pharmacy staff may perform inventory and fulfilment actions.
- BR-03: Only administrators may modify priority-rule configuration.
- BR-04: A booking requiring a prescription shall not be fulfilled unless the required prescription condition is satisfied.
- BR-05: Priority shall be calculated using the approved project policy; the exact weights must be documented in the design document and remain consistent during a calculation.
- BR-06: Availability shall be based on allocatable stock, not merely total physical stock, when stock is already reserved.
- BR-07: Equal-priority bookings shall use the configured tie-breaker, with booking timestamp as the default.
- BR-08: Cancellation shall release any reserved stock that has not been fulfilled.
- BR-09: Audit records shall not be deleted through ordinary user operations.
- BR-10: Patient history shall be limited to information necessary for the approved prioritization rule and shall not be used for unrelated purposes.

7. Other Requirements

- Database requirements: persistent storage shall maintain referential integrity between users, medicines, prescriptions, bookings, waitlist entries, inventory records, notifications and audit records.
- Backup requirement: project deployment shall have a database backup/recovery procedure appropriate to the selected environment.
- Internationalization: the interface shall be designed so that labels and notifications can be extended to additional languages if required.
- Logging: application errors and important business events shall be logged with timestamps and correlation/reference identifiers where practical.
- Compliance: the implementation team shall identify and follow applicable organizational and legal requirements for handling prescription and user information.

Appendix A: Glossary

Companion workbook: Project47_Field_Layouts_and_Report_Requirements.xlsx. The workbook contains field properties/attributes, validation rules, requirement mapping and report requirements.

Term	Definition
Booking	A customer request to obtain a specified quantity of a medicine.
Priority Score	A value calculated using configured factors to determine relative order of eligible bookings.
Allocation	Reservation/assignment of available medicine stock to a booking.
Waitlist	Ordered set of eligible bookings that cannot currently be fulfilled because sufficient stock is unavailable.
E-Prescription	Electronic prescription information issued through an authorized prescription workflow.
Available Stock	Quantity that is currently eligible for allocation after considering existing reservations.
Reserved Stock	Quantity set aside for an allocated booking but not yet fulfilled.

RBAC	Role-Based Access Control; authorization based on user role.
Audit Log	Record of important system actions and state changes for traceability.
Pharmacy Staff	Authorized user who manages inventory and booking fulfilment.
Administrator	Authorized user who manages system configuration, roles and reports.

Appendix B: Field Layouts

The following fields are proposed for the project database/UI. Final field lengths and database types may be adjusted during design. The companion Excel workbook provides the complete field layouts, validation rules, examples, report requirements and requirement mapping. In case of presentation differences, the Excel workbook and this appendix shall be maintained consistently.

B.1 Customer/User Registration

Field	Length	Data Type	Description	Mandatory
User ID	36	UUID/String	Unique user identifier	Y
Full Name	100	String	Customer/staff name	Y
Email	150	String	Login/contact email	Y
Phone	15	String	Contact number	Y
Password Hash	255	String	Securely stored password hash	Y
Role	30	Enum/String	Customer/Pharmacy Staff/Administrator	Y
Account Status	20	Enum/String	Active/Inactive/Locked	Y

B.2 Medicine

Field	Length	Data Type	Description	Mandatory
Medicine ID	36	UUID/String	Unique medicine identifier	Y
Medicine Name	150	String	Medicine name	Y
Category	80	String	Medicine category	N
Strength	50	String	Strength/form information	N
Prescription Required	1	Boolean	Whether prescription is required	Y
Status	20	Enum/String	Active/Inactive	Y

B.3 Inventory

Field	Length	Data Type	Description	Mandatory
Inventory ID	36	UUID/String	Unique inventory record identifier	Y
Medicine ID	36	UUID/String	Reference to medicine	Y
Available Quantity	10	Integer	Quantity currently available for	Y

			allocation	
Reserved Quantity	10	Integer	Quantity reserved for active bookings	Y
Fulfilled Quantity	10	Integer	Quantity already fulfilled	Y
Last Updated At	—	Timestamp	Time of latest inventory update	Y

B.4 Booking

Field	Length	Data Type	Description	Mandatory
Booking ID	36	UUID/String	Unique booking identifier	Y
User ID	36	UUID/String	Customer reference	Y
Medicine ID	36	UUID/String	Medicine reference	Y
Prescription ID	36	UUID/String	Linked prescription, if applicable	N
Quantity	10	Integer	Requested quantity	Y
Urgency	20	Enum/String	Configured urgency level	Y
Priority Score	10,2	Decimal	Calculated priority value	Y
Status	30	Enum/String	Requested/Allocated/Waitlisted/Fulfilled/Cancelled/Rejected	Y
Created At	—	Timestamp	Booking creation time	Y
Allocated At	—	Timestamp	Time at which stock was successfully allocated	N

B.5 Waitlist Entry

Field	Length	Data Type	Description	Mandatory
Waitlist ID	36	UUID/String	Unique waitlist record	Y
Booking ID	36	UUID/String	Booking reference	Y
Priority Score	10,2	Decimal	Priority used for ordering	Y
Position	10	Integer	Current waitlist position	Y
Status	20	Enum/String	Active/Promoted/Cancelled/Fulfilled	Y
Created At	-	Timestamp	Waitlist creation time	Y

B.6 Notification

Field	Length	Data Type	Description	Mandatory
Notification ID	36	UUID/String	Unique notification identifier	Y
User ID	36	UUID/String	Recipient	Y
Event Type	50	String	Booking / Allocation / Waitlist / Prescription / Fulfilment / Inventory / Rejection / Cancellation	Y
Message	500	String	Notification content	Y
Channel	20	Enum/String	Push/SMS/Email/In-app	Y

Status	20	Enum/String	Pending/Sent/Delivered/Failed	Y
Created At	-	Timestamp	Creation time	Y

B.7 Prescription

Field	Length	Data Type	Description	Mandatory
Prescription ID	36	UUID/String	Unique prescription identifier	Y
User ID	36	UUID/String	User who submitted the prescription	Y
Prescription Reference	150	String	Reference or document identifier	Y
Prescription Status	20	Enum/String	Submitted/Under Review/Verified/Rejected	Y
Submitted At	—	Timestamp	Prescription submission time	Y
Verified At	—	Timestamp	Time of prescription verification	N

B.8 Priority Rule

Field	Length	Data Type	Description	Mandatory
Rule ID	36	UUID/String	Unique priority rule identifier	Y
Rule Name	100	String	Name of the priority rule	Y
Urgency Weight	10,2	Decimal	Weight assigned to urgency	Y
Availability Weight	10,2	Decimal	Weight assigned to medicine availability	Y
Patient History Weight	10,2	Decimal	Weight assigned to permitted patient-history factor	Y
Version	20	String	Rule version used for auditability	Y
Status	20	Enum/String	Active/Inactive	Y
Created At	—	Timestamp	Rule creation time	Y

B.9 Audit Log

Field	Length	Data Type	Description	Mandatory
Audit ID	36	UUID/String	Unique audit record identifier	Y
Actor ID	36	UUID/String	User or service performing the action	Y
Action	100	String	Action performed	Y
Entity Type	50	String	Type of affected record	Y
Entity ID	36	UUID/String	Identifier of affected record	N
Timestamp	—	Timestamp	Time of action	Y
Details	500	String	Relevant action	N

			details	

Appendix C: Requirement Traceability Matrix

Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
FR-AUTH-01	Customer registration	Architecture: Auth Service	Design: Auth Module	Planned – auth_service	Planned – TC-AUTH-01	Planned – ST-AUTH-01
FR-MED-02	Display medicine availability	Architecture: Inventory Service	Design: Medicine Search	Planned – medicine_controller	Planned – TC-MED-02	Planned – ST-MED-02
FR-PRES-01	Attach e-prescription	Architecture: Prescription Service	Design: Prescription Module	Planned – prescription_service	Planned – TC-PRES-01	Planned – ST-PRES-01
FR-PRI-02	Calculate booking priority	Architecture: Priority Engine	Design: Priority Calculator	Planned – priority_engine	Planned – TC-PRI-02	Planned – ST-PRI-02
FR-PRI-04	Allocate by priority order	Architecture: Allocation Service	Design: Allocation Workflow	Planned – allocation_service	Planned – TC-PRI-04	Planned – ST-PRI-04
FR-WAIT-02	Maintain waitlist position	Architecture: Waitlist Service	Design: Waitlist Module	Planned – waitlist_service	Planned – TC-WAIT-02	Planned – ST-WAIT-02
FR-NOT-02	Send status notifications	Architecture: Notification Service	Design: Notification Handler	Planned – notification_service	Planned – TC-NOT-02	Planned – ST-NOT-02
FR-INV-03	Prevent negative inventory	Architecture: Inventory Service	Design: Inventory Transaction	Planned – inventory_service	Planned – TC-INV-03	Planned – ST-INV-03
FR-ADM-03	Audit important changes	Architecture: Audit Service	Design: Audit Logger	Planned – audit_service	Planned – TC-ADM-03	Planned – ST-ADM-03
NFR-SEC-02	Role-based authorization	Architecture: Security Layer	Design: RBAC	Planned – auth_middleware	Planned – TC-SEC-02	Planned – ST-SEC-02
NFR-SAFE-03	Transaction-safe allocation	Architecture: Allocation/DB Layer	Design: Transaction Control	Planned – allocation_service	Planned – TC-SAFE-03	Planned – ST-SAFE-03

End of Software Requirements Specification